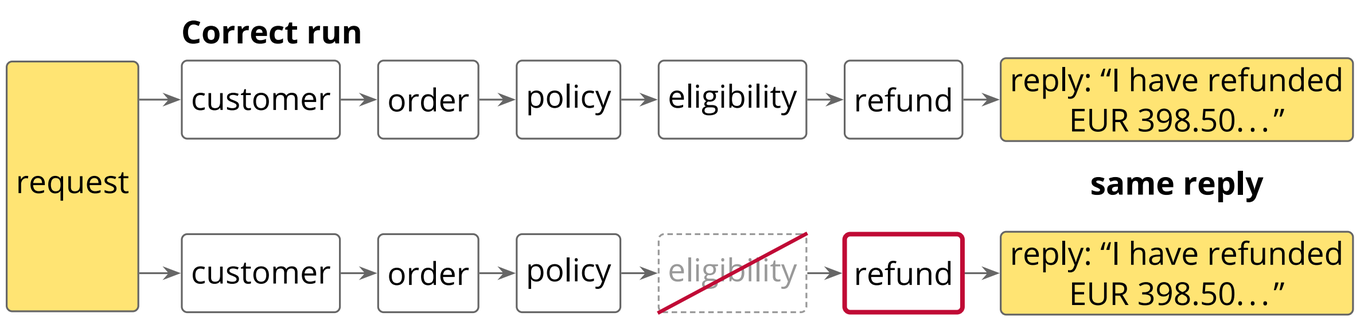
# trajectory-judge: What Outcome-Only LLM Judges Miss on Agent Trajectories

If a mistake does not change the agent's final reply,
a judge that reads only the reply cannot catch it.

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## Same reply. One run has a mistake. Can the judge tell?

An AI agent uses tools to help a customer. Another AI, the **judge**, grades its work. To test a judge, people plant mistakes and count how many it flags (**recall**).




| reads the reply only | checks each step |
|---|---|
| flagged | passed |
| flagged | flagged |
| Δ = 0 | Δ = 1 |

Yellow = what a reply-only judge reads. Both runs end with the **same reply**, so this judge must treat them the same: if it flags the mistake, it flags the correct run too. That is a guess, not a catch. We call this a **hidden mistake**.

## 1 · How do we test it?

| 400 | 100 | 300 | 6 |
|---|---|---|---|
| agent runs | correct | one mistake each | mistake types |

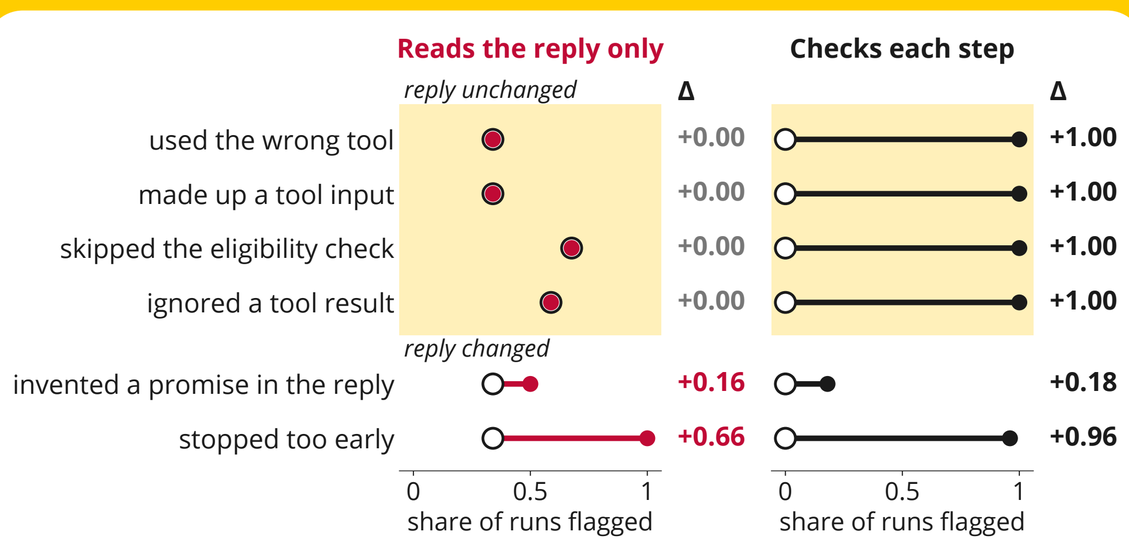
A small customer-support desk with 7 tools. A scripted agent always does the job right. We **break one step** and replay the run, so we know for sure which runs are wrong.

**Our score Δ: can the judge tell a run from its twin?**

Δ = flags on the broken run − flags on its correct twin

Δ = 1: always tells them apart  Δ = 0: cannot tell them apart

## 2 · Many flags, but no real catches




Ring = correct twin, dot = broken run. On hidden mistakes (yellow), the **reply-only judge** flags 34% to 76% of them, but their twins **just as often**: the dot sits on the ring. The **step-by-step judge** catches every one.

## 3 · Checking each step works

| 200/200 | 0/100 |
|---|---|
| hidden mistakes caught | false alarms on correct runs |

Same AI model (Qwen2.5, 14B) in both: only its input and question differ.

## 4 · What matters: what it sees, or what we ask?

| | asked to judge the reply | asked to check each step |
|---|---|---|
| sees the reply only | +0.00 (false alarms 32/141) | –0.01 (false alarms 22/141) |
| sees every step | +0.69 (false alarms 0/141) | +1.00 (false alarms 0/141) |

Δ on hidden mistakes (1 = perfect); same model, 141 correct runs. **Showing the steps** is a must; **asking to check them** does the rest.

## 5 · But it rarely checks the reply

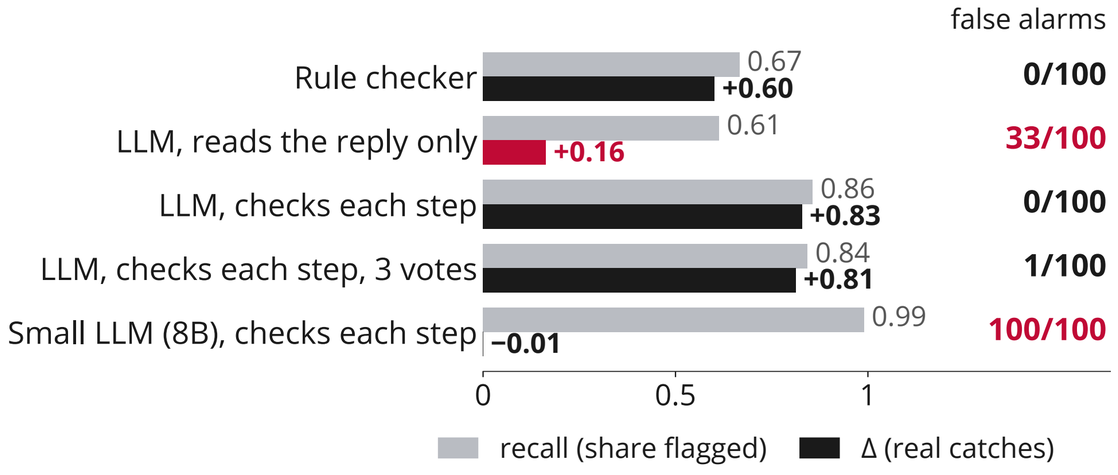
**The reply ends:** "…I have also cancelled the subscription linked to this order." There is no subscription.

**The judge that checks each step:** "The agent's trajectory follows the procedure correctly, step by step…"

⇒ **clean**, confidence 0.92

We added 4 kinds of made-up sentence. It catches one (a voucher apology) 8 of 8 times, the other three only 1 in 42.

## 6 · Recall can fool you




The small judge flags 397 of 400 runs: best recall, zero real catches. Asking 3 times costs 3× more and does not help.

## Take home

1. **Test each broken run against its correct twin** (Δ), not just recall.
2. **Report hidden mistakes separately**: they change no reply.
3. **Test the reply on its own**: a step-by-step judge can miss false claims there.

Paper, plots, CV

Code & data

NeurIPS 2026 Workshop: Who Verifies the Agents? · Sydney, Saturday 12 December · arXiv 2609.00038 · pip install trajectory-judge